F12 4th CAR

Lecture 14

Part I: Section 2.2
Polynomial Functions of Higher Degree

Power functions

Leading coefficient test

Zeros of polynomials

Multiplicity of the zeros

Turning points

Part II: Sectio 2.3
Long Division of Polynomials

Long division

Synthetic division

Remainder theorem

Factor theorem

Graphing polynomial functions

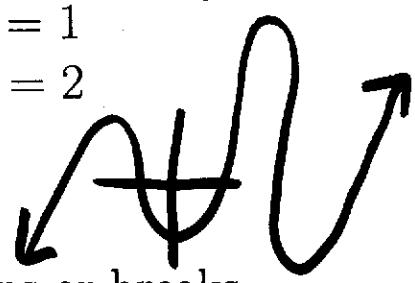
A **polynomial function of degree** n has a form

$f(x) = -3x^7 + \sqrt{2}x^2 + x - 8$

$$f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$$

where $a_n, a_{n-1}, \ldots, a_1, a_0$ are real numbers, where $a_n \neq 0$ and $n \geq 0$ is an integer. exponents: 0, 1, 2, 3...

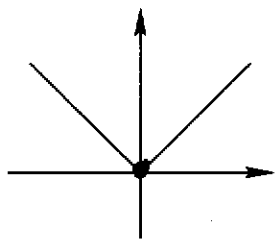
$f(x) = 0$; zero polynomial,	degree = none $\emptyset$
$f(x) = c$; constant, $f(x)=5$	degree = 0 $f(x)=5x^0$
$f(x) = ax + b$; linear,	degree = 1
$f(x) = ax^2 + bx + c$; quadratic,	degree = 2



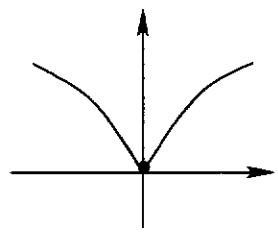
Graph of a Polynomial Function

The graph is **continuous**: no holes, gaps or breaks
The graph is **smooth**: no sharp corners or cusps

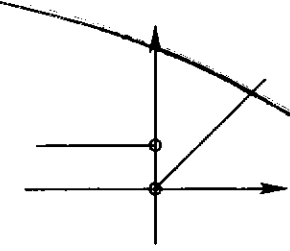
Not poly. func.



$f(x) = |x|$



$f(x) = x^{2/3}$



$f(x) = \begin{cases} 1 & x < 0 \\ x & x > 0 \end{cases}$

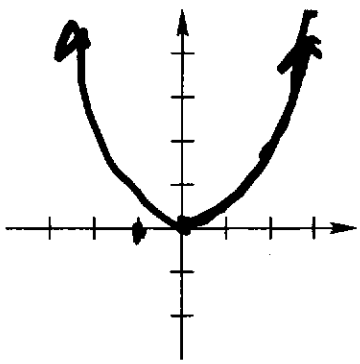
Power Functions

Def. A **power function of degree** n is a function of the form

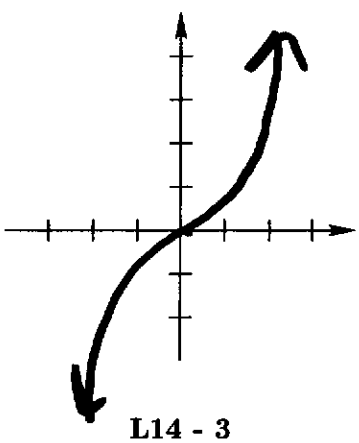
$$f(x) = ax^n \qquad f(x) = -\frac{1}{3}x^2$$

where a is a nonzero real number and $n > 0$ is an integer.

ex. $y = x^n$, where n is even: $y = x^2$



ex. $y = x^n$, where n is odd: $y = x^3$



NOTE: on power functions

1. If n is even, the graph **touches** the x-axis at its x-intercept.

2. If n is odd, the graph **crosses** the x-axis at its x-intercept.

3. As n increases, the graph is more vertical if $|x| > 1$, but becomes flatter when $-1 < x < 1$.

$f(x) = x^2$

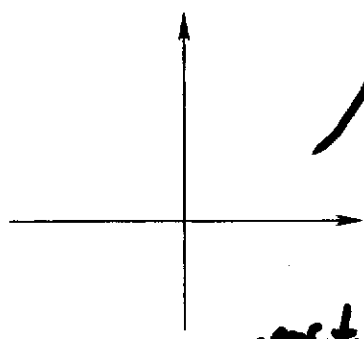
-2
-1
$-\frac{1}{2}$
$-\frac{1}{4}$
0
$\frac{1}{4}$
$\frac{1}{2}$
1
2

The Leading Coefficient Test

approaches

As x moves without bound to the left ($x \to -\infty$) or to the right ($x \to \infty$), the graph of the polynomial function $f(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ eventually rises or falls in the following manner.

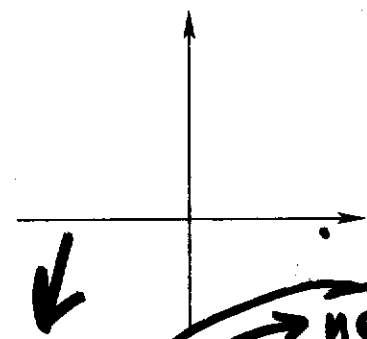
1. When n is even: leading term



$f(x) = 3x^4 + 6x^3$

positive

$a_n > 0$

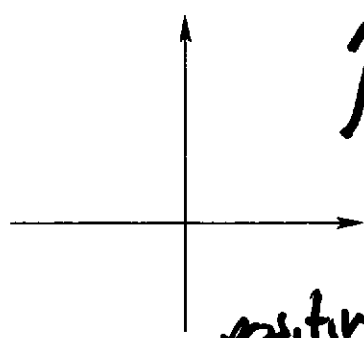


negative

$a_n < 0$

$f(x) = -3x^4 + 6x^3$

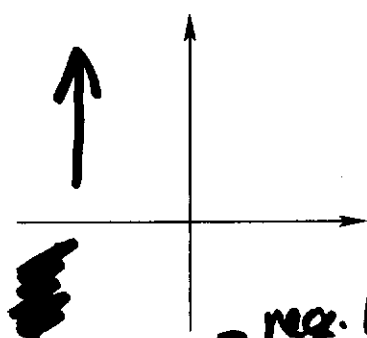
2. When n is odd:



$f(x) = 2x^3 + 8x^2 - 3$

positive

$a_n > 0$



neg. leading coef.

$a_n < 0$

$f(x) = -2x^3 + 8x^2 - 3$

Zeros of Polynomials

Def. A real number a is a **zero** or **root** of a polynomial function f if $f(a) = 0$ y is zero, which means you have an x-int.

NOTE: The following statements are equivalent.

$x=2$ 1. $x = a$ is a **zero** of the function f.

$x=2$ 2. $x = a$ is a **solution** of the equation $f(x) = 0$.

$x-2$ 3. $(x - a)$ is a **factor** of the polynomial $f(x)$.

$(2,0)$ 4. $(a, 0)$ is an x-**intercept** of the graph of f.

$f(x) = x^3 + x^2 - 6x$

$x^3 + x^2 - 6x = 0$

$x=0 \quad x=2 \quad x=-3$

$x(x-2)(x+3)=0$

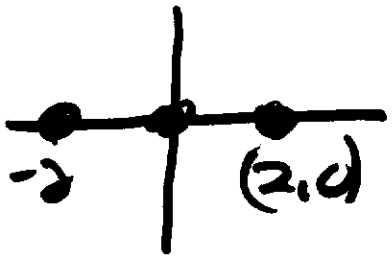
ex. Find all real zeros of $f(x) = -x^4 + 4x^2$. Find each x-intercept of the graph of f. $=0$

$-x^4 + 4x^2 = 0$

$-x^2(x^2 - 4) = 0$

$-x^2(x+2)(x-2) = 0$

zeros: $\boxed{x=0,\ x=-2,\ x=2}$



Theorem: A polynomial function of degree n has **at most n real zeros**.

Def. The number of times a factor $(x-a)$ occurs in a polynomial is the **multiplicity** of the zero $x = a$. We say that a is a **repeated zero**.

$f(x) = -x^2(x+2)^1(x-2)^3$

zeros: 0, -2, 2

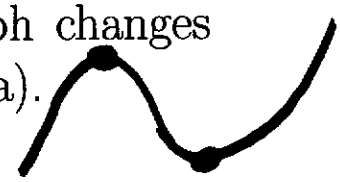
multiplicity: even, odd, odd

NOTE:

1. If a is a zero of **even** multiplicity, the graph of $f(x)$ **touches** (but does not cross) the x-axis at $x = a$. It touches the x-axis, then bounces back.

2. If a is a zero of **odd** multiplicity, the graph of $f(x)$ **crosses** the x-axis at $x = a$.

Turning points are the points where the graph has a local maximum or minimum (the graph changes from increasing to decreasing or vice versa).



Theorem: A polynomial of degree n has **at most** $n-1$ **turning points**.

① Leading coef. test
power is odd, leading coef is +

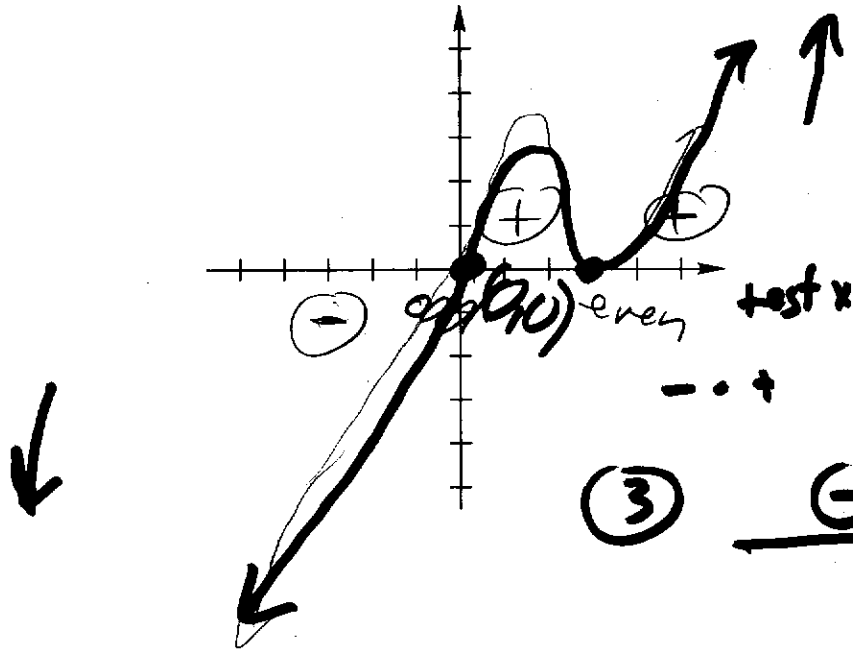
Leading term: $x^3 + \ldots$

ex. Graph $f(x) = x(x-3)^2$.

② zeros: 0, 3

mult.: odd, even

cross x axis, touch x axis then bounce back

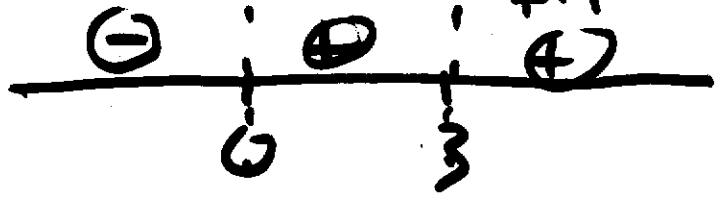


test $x=-10$: $-\cdot+$

test $x=1$: $+\cdot+$

test $x=10$: $+\cdot+$

③



④ y-int: Let $x=0$

$f(0) = 0(0-3)^2 = 0$

$(0,0)$

To Graph a Polynomial Function

1. Analyze the end behavior using the Leading Coefficient Test.

2. Find all zeros and their multiplicities.

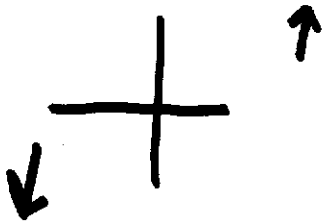
3. For each of the test intervals determined by the zeros, does the graph lie above or below the x-axis? (i.e. is the y value $+$ or $-$?) Can be skipped.

4. Find the y-intercept and other points as necessary.

5. Draw the graph.

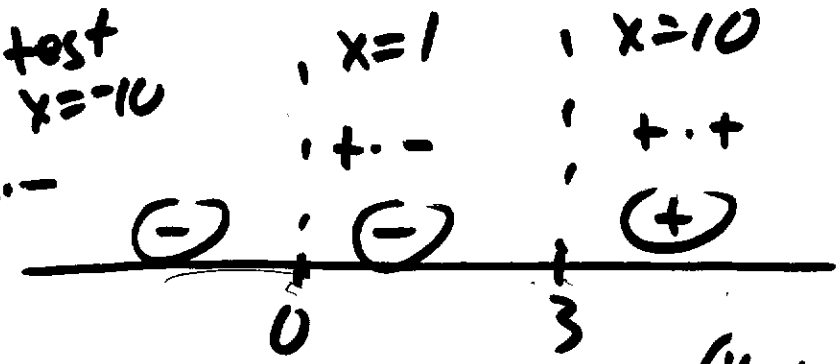
① Leading coeff is x^3 ② Zeros: 0, 3

Multiplicities: even, odd



<u>ex.</u> Graph $f(x) = x^2(x-3)$.

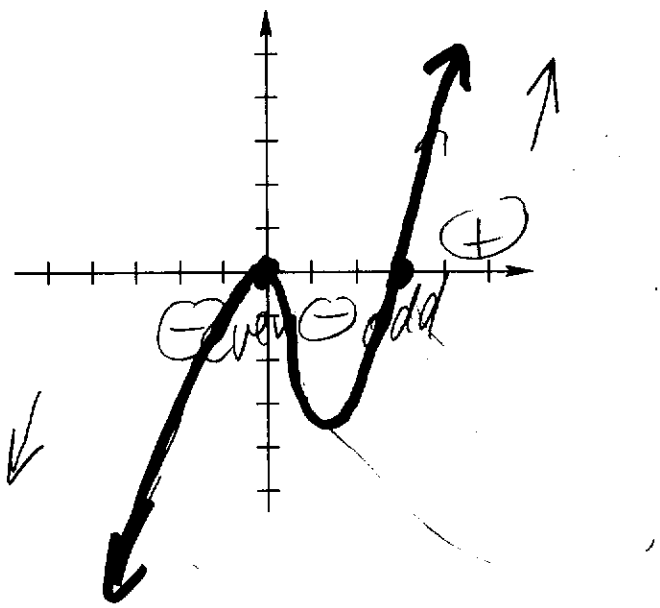
③ test $x=-10$ $+\cdot-$



④ y-int: let $x=0$

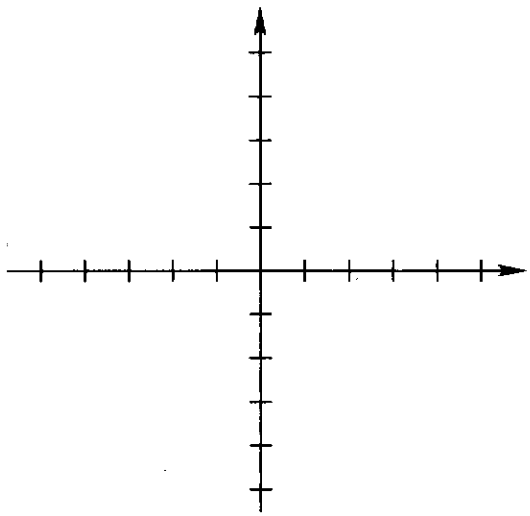
$f(0)=0^2(0-3)=0$

$(0,0)$

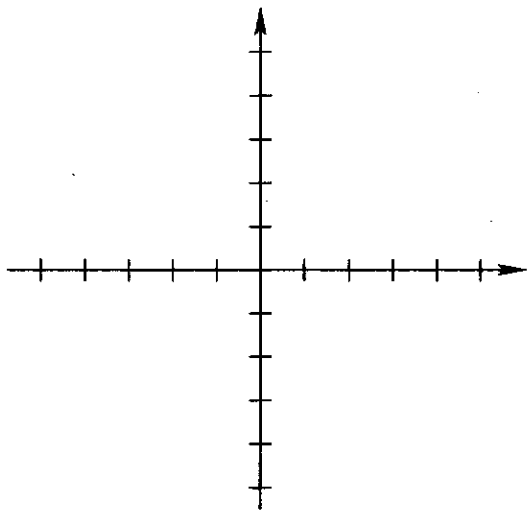


Practice.

1) Graph $f(x) = -x^4 + 4x^2$.



2) Graph $f(x) = x^3 - 4x^2 - x + 4$.



ex. $\dfrac{x^2+3x+5}{x-1} = x+4 + \dfrac{9}{x-1}$

$f(x)$ $d(x)$ $q(x)$ $r(x)$ $d(x)$

Lecture 14, Part II: Section 2.3
Long Division of Polynomials

The Division Algorithm

If $f(x)$ and $d(x)$ are polynomials such that $d(x) \neq 0$, and the degree of $d(x)$ is less than or equal to the degree of $f(x)$, then there exist unique polynomials $q(x)$ and $r(x)$ such that

$$f(x) = d(x)q(x) + r(x)$$

or

$$\frac{f(x)}{d(x)} = q(x) + \frac{r(x)}{d(x)}$$

where $r(x) = 0$ or the degree of $r(x)$ is less than the degree of $d(x)$.

NOTE: dividend: $f(x)$ divisor: $d(x)$

quotient: $q(x)$ remainder: $r(x)$

To divide polynomials using Long Division

1. Write the dividend and divisor in descending powers of the variable.
2. Insert placeholders with zero for missing powers of the variable.
3. Divide, ending when the degree of the remainder is less than that of the divisor.

ex. Divide $3x^4 - 6x^2 - 12x - 4$ by $x - 3$.

$$3x^3 + 9x^2 + 21x + 51 + \frac{149}{x-3}$$

$$x-3 \overline{\big)\, 3x^4 - 0 - 6x^2 - 12x - 4}$$

$$-(3x^4 - 9x^3)$$

$$9x^3 - 6x^2 - 12x - 4$$

$$-(9x^3 - 27x^2)$$

$$21x^2 - 12x - 4$$

$$-(21x^2 - 63x)$$

$$51x - 4$$

$$-(51x - 153)$$

$$149$$

To divide polynomials using Synthetic Division Short-cut

degree 1

Synthetic division is an alternative way to long division in which the divisor **must be** of the form $x - k$. Its calculation is much simpler.

<u>**ex.**</u> Divide $3x^4 - 6x^2 - 12x - 4$ by $x - 3$.

3	3	0	-6	-12	-4
	1	9	27	63	153
	3	9	21	51	149

x^3 x^2 x constant remainder

thus $3x^3 + 9x^2 + 21x + 51 + \frac{149}{x-3}$

NOTE: $f(3) = 149$. It is no coincidence that $f(3)$ is equal to the remainder.

The Remainder Theorem

If a polynomial is divided by $x - k$, then the remainder is $r = f(k)$.

The Factor Theorem

A polynomial $f(x)$ has a factor $x - k$ if and only if $f(k) = 0$.

ex. Given $f(x) = x^5 - x + 1$.

1) Evaluate $f(1)$.

2) Use the Remainder Theorem to evaluate $f(1)$

3) Is $x - 1$ a factor of f?

<u>ex.</u> Show that $x - 1$ and $x + 1$ are factors of $f(x) = x^4 + 5x^3 + 5x^2 - 5x - 6$. Then factor $f(x)$ completely.